

Microelectronics

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1 Microelectronics

I Signal

i Signal

A signal is a function that conveys information about a phenomenon. Signal processing is usually most conveniently performed by electronic systems on voltage or current signals, especially the former. Only voltage and current signals are studied here.

ii Signal source

A signal source is typically represented in the Thévenin form, a voltage source in a series connection with a resistance, or the Norton form, a current source in a parallel connection with a resistance.

iii Cascade

Output of a component as the input of another.

iv Transducer

A transducer is a device that convert a signal to a voltage or current signal or vice versa.

v Signal processing

Signal processing includes analog signal processing (ASP), analog-to-digital converter (ADC), digital signal processor (DSP), digital-to-analog converter (DAC), amplifier, etc.

vi Coupling capacitor

A capacitor connected in series between stages to pass AC signal while blocking DC.

vii Symbol convention

- Lowercase character with uppercase subscript means total signal, e.g. $i_C(t)$.
- Lowercase character with lowercase subscript means AC signal, e.g. $i_c(t)$.
- Uppercase character with uppercase subscript means DC signal, e.g. I_C .
- Uppercase character with lowercase subscript means peak signal, e.g. I_c .

Their relation is

$$i_C(t) = i_c(t) + I_C$$

$$i_c(t) = I_c \sin(\omega t + \varphi)$$

viii differential and common-mode signal

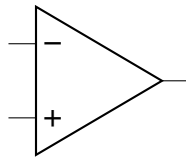
For a component with two input signals, the differential signal (AC signal) v_d is the difference between the two input signals, and the common-mode signal (DC signal) v_{cm} is the algebraic average of the two input signals.

II Amplifier (amp)

i Amplifier

An electronic device which amplifies an electrical signal by a scale called amplifier gain A and optionally convert voltage signal to current signal or vice versa.

ii Diagram



iii Two-port model

An amplifier can be modeled as a two port model, where an one input port and one output port is related by the amplifier, which acts as a controlled source controlled by the input port voltage or current which enforces voltage or current on the output port.

iv Circuit model of voltage amplifier

A voltage amplifier draws current from a source v_s with internal resistance R_s with input resistance R_i and voltage v_{in} across it and supplies $v_c = \mu v_{in}$ to a load network with output resistance R_o and load R_L , where μ , also denoted A_{vo} , is open-loop voltage gain factor.

$$\frac{v_{in}}{v_s} = \frac{R_i}{R_s + R_i}$$

$$\frac{v_c}{v_{in}} = \mu$$

$$\frac{v_{out}}{v_c} = \frac{R_L}{R_o + R_L}$$

Voltage gain:

$$A_v = \frac{v_{out}}{v_{in}} = \mu \frac{R_i R_L}{(R_s + R_i)(R_o + R_L)}$$

$|A| < |\mu|$ due to loading effects.

A good amplifier requires small loading effect:

$$R_i \gg R_s, \quad R_o \ll R_L, \quad A_v \approx \mu$$

v Circuit model of current amplifier

A current amplifier draws current from a current source i_s with internal resistance R_s in parallel with input resistance R_i and current i_{in} through it and supplies $i_c = \beta i_{in}$ by CCCS to a load network with output resistance R_o in parallel and load R_L , where β , also denoted A_{cs} , is short-circuit current gain factor.

$$\begin{aligned}\frac{i_{in}}{i_s} &= \frac{R_s}{R_s + R_i} \\ \frac{i_c}{i_{in}} &= \beta \\ \frac{i_{out}}{i_c} &= \frac{R_o}{R_o + R_L}\end{aligned}$$

Current gain:

$$A_c = \frac{i_{out}}{i_{in}} = \beta \frac{R_s R_o}{(R_s + R_i)(R_o + R_L)}$$

$|A| < |\beta|$ due to loading effects.

A good amplifier requires small loading effect:

$$R_i \ll R_s, \quad R_o \gg R_L, \quad A_c \approx \beta$$

vi Circuit model of transconductance amplifier

A transconductance amplifier draws current from a voltage source v_s with internal resistance R_s in series with input resistance R_i and current i_{in} through it and supplies $i_c = g v_{in}$ by VCCS to a load network with output resistance R_o in parallel and load R_L , where μ is open-loop transconductance factor.

$$\begin{aligned}\frac{v_{in}}{v_s} &= \frac{R_i}{R_s + R_i} \\ \frac{i_c}{v_{in}} &= g \\ \frac{i_{out}}{i_c} &= \frac{R_o}{R_o + R_L}\end{aligned}$$

Gain:

$$A_g = \frac{i_{out}}{v_{in}} = g \frac{R_i R_o}{(R_s + R_i)(R_o + R_L)}$$

$|A| < |g|$ due to loading effects.

A good amplifier requires small loading effect:

$$R_i \gg R_s, \quad R_o \gg R_L, \quad A_g \approx g$$

vii Circuit model of transresistance amplifier

A transresistance amplifier draws current from a current source i_s with internal resistance R_s in parallel with input resistance R_i and current i_{in} through it and supplies $v_c = r i_{in}$ by CCVS to a load network with output resistance R_o in series and load R_L , where r is short-circuit transresistance factor.

$$\frac{i_{in}}{i_s} = \frac{R_s}{R_s + R_i}$$

$$\frac{v_c}{i_{in}} = r$$

$$\frac{v_{out}}{v_c} = \frac{R_L}{R_o + R_L}$$

Gain:

$$A_r = \frac{v_{out}}{i_{in}} = \beta \frac{R_s R_L}{(R_s + R_i)(R_o + R_L)}$$

$|A| < |r|$ due to loading effects.

A good amplifier requires small loading effect:

$$R_i \ll R_s, \quad R_o \ll R_L, \quad A_c \approx r$$

viii Amplifier gain in decibels (dB)

A power gain A_p in dB is defined as

$$G_p = 10 \log_{10}(A_p)$$

A voltage gain A_v in dB is defined as

$$G_v = 20 \log_{10}(A_v)$$

A current gain A_c in dB is defined as

$$G_c = 20 \log_{10}(A_c)$$

ix Saturation

The linear characteristic only remains within a range. A positive and negative saturation levels can be observed.

x S parameter

If the energy enters through port i and emerges from port j , the S parameter is named S_{ji} .

xi Unilateral amplifier

An unilateral amplifier is an amplifier in which the signal transfer is effectively one-way, that is, the output signal is only dependent on the input signal but not what is connected to the output terminal. That is for each output port j and each input port i , $S_{ij} = 0$.

xii Power supply

To amplify signal, power supplies are needed, but often omitted in the diagram.

- V_{CC+} or V_{CC} : positive supply voltage, denoted on the + side of an op-amp or omitted;
- V_{CC-} or $-V_{EE}$: negative supply voltage, denoted on the – side of an op-amp or omitted.

The DC power delivered is the sum of power from the positive and negative power supply terminals V_{CC} and V_{EE}

$$P_{DC} = V_{CC}I_{CC} + V_{EE}I_{EE}$$

Suppose input and output signals have power P_I and P_L .

$$P_{DC} + P_I = P_L + P_{\text{dissipated}}$$

where $P_{\text{dissipated}}$ is the power dissipated.

xiii Classification based on frequency response

A band-pass amplifier is also called tuned amplifier. A low-pass amplifier is also called DC amplifier. A amplifier directly coupled with a coupling capacitor, called capacitively coupled, is called AC amplifier.

III Operational Amplifier (op amp or op-amp)

i Definition

An op amp has five terminals:

- v_+ : non-inverting input voltage of the terminal at the + sign;
- v_- : inverting input voltage of the terminal at the – sign;
- v_o : output voltage of the terminal at the opposite side of inputs;
- V_{CC+} or V_{CC} : positive supply voltage, denoted on the + side of an op-amp or omitted;
- V_{CC-} or $-V_{EE}$: negative supply voltage, denoted on the – side of an op-amp or omitted.

$$v_o = A(v_+ - v_- - v_{os})$$

where A is open-loop gain or open-circuit gain, and v_{os} is input offset voltage.

$$v_{ld} = v_+ - v_-$$

$$v_{lcm} = \frac{v_+ + v_-}{2}$$

ii Ideal op amp

- Infinite input impedance: No signal current flowing into v_+ and v_- .
- Infinite open-loop gain
- Infinite bandwidth: open-loop gain remains constant in all frequencies.
- Zero output impedance: The amplifier is unilateral.
- Zero input offset voltage: $v_{os} = 0$, which implies zero common-mode gain, i.e., infinite common-mode rejection.

This is a linear element with inputs v_+ and v_- and output v_o .

We can construct a virtual short circuit between v_+ and v_- for ideal op amp.

Note that while $v_+ - v_-$, v_o may be any voltage since $A = \infty$ and $0 * \infty$ can be any number in $[0, \infty]$ in limit sense.

iii Feedback configuration

To have stable signal, op amp in a circuit is connected with a negative feedback configuration, which consists of the op amp and a connection from v_o to either of the input terminals with a resistance on it, called feedback connection. When the feedback connection connects to v_- , it is called inverting configuration. When the feedback connection connects to v_+ , it is called non-inverting configuration.

iv Inverting amplifier

An inverting amplifier consists of an op amp where input voltage v_i is connected to v_- through an input resistance R_i (current from v_i to v_-), v_+ is connected to ground, and v_o is fed back to v_- through a feedback resistance R_f (current from v_- to v_o).

$$\begin{aligned} v_+ &= v_- = 0 \\ i_i &= \frac{v_i}{R_i} = i_f = -\frac{v_o}{R_f} \\ v_o &= -\frac{R_f}{R_i} v_i \end{aligned}$$

The closed-loop gain G is defined as

$$G = \frac{v_o}{v_i} = -\frac{R_f}{R_i}$$

When $-G < 1$, it's called an attenuator.

Effect of finite open-loop gain:

$$\begin{aligned} v_+ &= 0 \\ v_o &= -A v_- \\ i_i &= \frac{v_i - v_-}{R_i} = \frac{v_i + \frac{v_o}{A}}{R_i} = i_f = -\frac{v_o + \frac{v_o}{A}}{R_f} \\ G &= \frac{v_o}{v_i} = -\frac{R_f}{R_i \left(1 + \frac{1 + \frac{R_f}{R_i}}{A} \right)} = \frac{G_{\text{ideal}}}{1 + \frac{1 - G_{\text{ideal}}}{A}} \\ \lim_{A \rightarrow \infty} G &= -\frac{R_f}{R_i} = G_{\text{ideal}} \end{aligned}$$

v Transresistance amplifier

A transresistance amplifier consists of an op amp where input current i_i is connected to v_- , v_+ is connected to ground, and v_o is fed back to v_- through two resistances R_o (current from x to v_o) and R_f (current from v_- to x) in series connection, with the node x between R_o and R_f connected to ground through a resistance R_g (current from x to ground).

$$v_x = -R_f i_i$$

$$i_g = -\frac{R_f}{R_g} i_i$$

$$i_o = \left(1 + \frac{R_f}{R_g}\right) i_i$$

$$v_o = -R_f i_i - R_o i_o = -\left(R_f + R_o + \frac{R_f R_o}{R_g}\right) i_i$$

$$\frac{v_o}{i_i} = -\left(R_f + R_o + \frac{R_f R_o}{R_g}\right)$$

vi Weighted summing amplifier

A weighted summing amplifier consists of an op amp, input voltages $v_1, v_2, \dots, v_n$, n input resistors $R_1, R_2, \dots, R_n$, and a feedback resistor R_f , where for any $k \in \mathbb{N} \wedge k \leq n$, v_k is connected to v_- through R_k , v_+ is connected to ground, and v_o is fed back to v_- through R_f .

$$i_k = \frac{v_k}{R_k}$$

$$\sum_{k=1}^n i_k = i_f = \frac{-v_o}{R_f}$$

$$v_o = -R_f \sum_{k=1}^n \frac{v_k}{R_k}$$

When $R_k = R$ for all $k \in \mathbb{N} \wedge k \leq n$:

$$v_o = -\frac{R_f}{R} \sum_{k=1}^n v_k$$

vii Non-inverting amplifier

A non-inverting (or noninverting) amplifier consists of an op amp where input voltage v_i is connected to v_+ , v_- is connected ground through a ground resistance R_g (current from v_- to ground), and v_o is fed back to v_- through a feedback resistance R_f (current from v_o to v_-).

$$v_+ = v_- = v_i$$

$$i_g = \frac{v_i}{R_g} = i_f = \frac{v_o - v_i}{R_f}$$

$$v_o = \left(1 + \frac{R_f}{R_g}\right) v_i$$

The closed-loop gain G is defined as

$$G = \frac{v_o}{v_i} = 1 + \frac{R_f}{R_g}$$

Effect of finite open-loop gain:

$$\begin{aligned} v_+ &= v_i \\ v_o &= A(v_i - v_-) \\ v_- &= v_i - \frac{v_o}{A} \\ i_g = \frac{v_-}{R_g} &= \frac{v_i - \frac{v_o}{A}}{R_g} = i_f = \frac{v_o - v_-}{R_f} = \frac{v_o - v_i + \frac{v_o}{A}}{R_f} \\ G = \frac{v_o}{v_i} &= \frac{1 + \frac{R_f}{R_g}}{1 + \frac{1 + \frac{R_f}{R_g}}{A}} = \frac{G_{\text{ideal}}}{1 + \frac{G_{\text{ideal}}}{A}} \\ \lim_{A \rightarrow \infty} G &= 1 + \frac{R_f}{R_g} = G_{\text{ideal}} \end{aligned}$$

viii Unity gain amplifier, unity gain buffer, buffer amplifier, or voltage follower

A unity gain amplifier consist of an op amp where input voltage v_i is connected to v_+ , and v_o is fed back to v_- .

$$v_o = v_- = v_+ = v_i$$

ix Difference amplifier or differential amplifier

A difference amplifier consists of an op amp where first input voltage v_1 is connected to v_- through an input resistance R_1 (current from v_1 to v_-), second input voltage v_2 is connected to v_+ through an input resistance R_2 (current from v_2 to v_+), v_+ is connected to ground through a ground resistance R_g (current from v_+ to ground), and v_o is fed back to v_- through a feedback resistance R_f (current from v_- to v_o). Denote common-mode signal $v_{cm} = \frac{v_1 + v_2}{2}$ and differential signal $v_d = v_2 - v_1$.

$$\begin{aligned} v_o &= -\frac{R_f}{R_1}v_1 + \frac{R_g}{R_2 + R_g} \left(1 + \frac{R_f}{R_1} \right) v_2 \\ i_1 &= \frac{1}{R_1} \left(v_1 - \frac{R_g}{R_2 + R_g} v_2 \right) \\ i_2 &= \frac{v_2}{R_2 + R_g} \end{aligned}$$

Common-mode gain A_{cm} and differential gain A_d , defined by

$$v_o = v_{cm}A_{cm} + v_dA_d,$$

are

$$A_{cm} = \frac{R_g}{R_2 + R_g} \left(1 - \frac{R_2}{R_g} \frac{R_f}{R_1} \right)$$

$$A_d = \frac{R_g}{R_2 + R_g} \left(1 + \frac{R_2 + 2R_g}{R_g} \frac{R_f}{R_1} \right)$$

Common-mode rejection ratio (CMRR) is defined as

$$\text{CMRR} = 20 \log \left| \frac{A_d}{A_{cm}} \right|$$

To make $A_{cm} = 0$, we need

$$\frac{R_f}{R_1} = \frac{R_g}{R_2},$$

at which

$$A_d = \frac{R_f}{R_1} = \frac{R_g}{R_2}$$

Differential input resistance R_d , defined as, when we connect v_1 and v_2 to a source v_d ($v_d = v_2 - v_1$ and $i_d = i_2 = -i_1$),

$$\frac{v_d}{i_d},$$

is

$$R_d = R_1 + R_2$$

Thus, high differential gain hurts differential input resistance, which lower the voltage applied to the amplifier since source has internal resistance.

Proof. Use superposition. With v_1 enabled and v_2 disabled,

$$v_+ = v_- = 0$$

$$i_2 = i_g = 0$$

$$i_1 = \frac{v_1}{R_1} = i_f$$

$$v_o = -i_f R_f = -\frac{R_f}{R_1} v_1$$

With v_2 enabled and v_1 disabled,

$$i_2 = i_g = \frac{v_2}{R_2 + R_g}$$

$$v_+ = v_- = i_g R_g = \frac{R_g}{R_2 + R_g} v_2$$

$$i_1 = -\frac{v_-}{R_1} = -\frac{R_g}{R_1(R_2 + R_g)} v_2 = i_f$$

$$v_o = v_- - i_f R_f = \frac{R_g}{R_2 + R_g} v_2 + \frac{R_g R_f}{R_1(R_2 + R_g)} v_2 = \frac{R_g}{R_2 + R_g} \left(1 + \frac{R_f}{R_1} \right) v_2$$

Sum:

$$v_o = -\frac{R_f}{R_1} v_1 + \frac{R_g}{R_2 + R_g} \left(1 + \frac{R_f}{R_1} \right) v_2$$

$$i_1 = \frac{v_1}{R_1} - \frac{R_g}{R_1(R_2 + R_g)} v_2 = \frac{1}{R_1} \left(v_1 - \frac{R_g}{R_2 + R_g} v_2 \right)$$

$$i_2 = \frac{v_2}{R_2 + R_g}$$

$$v_1 = v_{cm} - \frac{v_d}{2}$$

$$v_2 = v_{cm} + \frac{v_d}{2}$$

$$\begin{aligned} v_o &= -\frac{R_f}{R_1} \left(v_{cm} - \frac{v_d}{2} \right) + \frac{R_g}{R_2 + R_g} \left(1 + \frac{R_f}{R_1} \right) \left(v_{cm} + \frac{v_d}{2} \right) \\ &= \frac{R_g}{R_2 + R_g} \left(1 - \left(\frac{R_2 + R_g}{R_g} - 1 \right) \frac{R_f}{R_1} \right) v_{cm} + \frac{R_g}{R_2 + R_g} \left(1 + \left(\frac{R_2 + R_g}{R_g} + 1 \right) \frac{R_f}{R_1} \right) v_d \\ &= \frac{R_g}{R_2 + R_g} \left(1 - \frac{R_2 R_f}{R_g R_1} \right) v_{cm} + \frac{R_g}{R_2 + R_g} \left(1 + \frac{R_2 + 2R_g}{R_g} \frac{R_f}{R_1} \right) v_d \end{aligned}$$

$$A_{cm} = \frac{R_g}{R_2 + R_g} \left(1 - \frac{R_2 R_f}{R_g R_1} \right)$$

$$A_d = \frac{R_g}{R_2 + R_g} \left(1 + \frac{R_2 + 2R_g}{R_g} \frac{R_f}{R_1} \right)$$

To make $A_{cm} = 0$, we need

$$1 = \frac{R_2 R_f}{R_g R_1}$$

$$\frac{R_f}{R_1} = \frac{R_g}{R_2}$$

at which

$$\begin{aligned} A_d &= \frac{R_g}{R_2 + R_g} \left(1 + \frac{R_2 + 2R_g}{R_g} \frac{R_f}{R_1} \right) \\ &= \frac{R_g}{R_2 + R_g} \left(1 + \frac{R_2 + 2R_g}{R_2} \right) \\ &= \frac{R_g}{R_2} \end{aligned}$$

Differential input resistance:

$$i_d = \frac{v_+}{R_g} = \frac{v_2 - v_+}{R_2} = \frac{v_+ - v_2 + v_d}{R_1}$$

$$R_2 v_+ = R_g v_2 - R_g v_+$$

$$R_1 v_2 - R_1 v_+ = R_2 v_+ - R_2 v_2 + R_2 v_d$$

$$v_2 = \left(\frac{R_2 + R_g}{R_g} \right) v_+$$

$$(R_1 + R_2)(v_2 - v_+) = R_2 v_d$$

$$v_+ = \frac{R_g}{R_1 + R_2} v_d$$

$$i_d = \frac{1}{R_1 + R_2} v_d$$

$$R_d = \frac{v_d}{i_d} = R_1 + R_2$$

□

x Instrumental amplifier

TODO

xi Inverting integrator

An inverting integrator consists of an op amp where input voltage v_i is connected to v_- through an input resistor R_i , v_+ is connected to ground, and v_o is fed back to v_- through a feedback capacitor C_f .

Time domain:

$$i_i = \frac{v_i}{R_i} = i_f = -C_f \frac{dv_o}{dt}$$
$$v_o = v_o(0^-) - \int_0^t \frac{v_i(\tau)}{R_i C_f} d\tau$$

Laplace domain:

$$i_i = \frac{v_i}{R_i} = i_f = -C_f s v_o + C_f v_o(0^-)$$
$$v_o = \frac{v_o(0^-)}{s} - \frac{v_i}{R_i C_f s}$$

xii inverting differentiator

An inverting differentiator consists of an op amp where input voltage v_i is connected to v_- through an input capacitor C_i , v_+ is connected to ground, and v_o is fed back to v_- through a feedback resistor R_f .

Time domain:

$$i_i = C_i \frac{dv_i}{dt} = i_f = \frac{-v_o}{R_f}$$
$$v_o = R_f C_i \left(v_i(0^-) - \int_0^t v_i(\tau) d\tau \right)$$

Laplace domain:

$$i_i = C_i s v_i - C_i v_i(0^-) = i_f = \frac{-v_o}{R_f}$$
$$v_o = R_f C_i (v_i(0^-) - s v_i)$$

xiii Active filter

Given a (passive or active) filter v_f of an input v_s and an inverting amplifier, non-inverting amplifier, or unit gain buffer enforcing $v_o = K v_i$, by connecting the ground and v_i of the amplifier to the two terminals of the v_f , the two terminals ground and v_o become an active filter of v_s enforcing $v_o = K v_f$.